

Table 15: The criteria induced from human judgments for validating the experiment designs, which are used to align model-based evaluations with actual human preferences.

Types	Criteria	Texts
Experiment	Clarity	1. The experiment design is extremely unclear, with critical details missing or ambiguous, making it nearly impossible for others to understand the setup, procedure, or expected outcomes. 2. The experiment design lacks significant clarity, with many important aspects poorly explained or omitted, challenging others to grasp the essential elements of the setup, procedure, or expected outcomes. 3. The experiment design is moderately clear, but some aspects are not detailed enough, leaving room for interpretation or confusion about the setup, procedure, or expected outcomes. 4. The experiment design is mostly clear, with most aspects well-described, allowing others to understand the setup, procedure, and expected outcomes with minimal ambiguity. 5. The experiment design is exceptionally clear, precise, and detailed, enabling easy understanding of the setup, procedure, and expected outcomes, with no ambiguity or need for further clarification.
		1. The experiment design demonstrates a fundamental misunderstanding of the research problem, lacks alignment with scientific methods, and shows no evidence of validity in addressing the research questions or testing the proposed methods. 2. The experiment design has significant flaws in its approach to the research problem and scientific method, with minimal or questionable evidence of validity, making it largely ineffective in addressing the research questions or testing the proposed methods. 3. The experiment design is generally aligned with the research problem and scientific method but has some limitations in its validity, offering moderate evidence that it can somewhat effectively address the research questions or test the proposed methods. 4. The experiment design is well-aligned with the research problem and scientific method, providing strong evidence of validity and effectively addressing the research questions and testing the proposed methods, despite minor limitations. 5. The experiment design excellently aligns with the research problem and scientific method, demonstrating robust evidence of validity and outstandingly addressing the research questions and testing the proposed methods without significant limitations.
	Robustness	1. The experiment design demonstrates a fundamental lack of understanding of the scientific method, with no evidence of durability or adaptability across varying conditions, leading to highly unreliable and non-replicable results. 2. The experiment design shows minimal consideration for robustness, with significant oversights in addressing variability and ensuring consistency across different scenarios, resulting in largely unreliable outcomes. 3. The experiment design adequately addresses some aspects of robustness but lacks comprehensive measures to ensure durability and consistency across a wide range of conditions, leading to moderate reliability. 4. The experiment design incorporates a solid understanding of robustness, with clear efforts to ensure the experiment's durability and consistency across diverse conditions, though minor improvements are still possible for optimal reliability. 5. The experiment design exemplifies an exceptional commitment to robustness, with meticulous attention to durability and adaptability across all possible conditions, ensuring highly reliable and universally applicable results.
		1. The experiment design is fundamentally unfeasible, with insurmountable resource, time, or technological constraints that make implementation virtually impossible within the proposed framework. 2. The experiment design faces significant feasibility challenges, including major resource, time, or technological limitations, that heavily compromise its practical execution and likelihood of success. 3. The experiment design is somewhat feasible, with moderate constraints on resources, time, or technology that could be addressed with adjustments, though these may not guarantee success. 4. The experiment design is largely feasible, with minor resource, time, or technological limitations that can be effectively managed or mitigated, ensuring a high probability of successful implementation. 5. The experiment design is highly feasible, with no significant constraints on resources, time, or technology, indicating that it can be implemented smoothly and successfully within the proposed framework.
	Feasibility	1. The experiment design lacks critical details, making it virtually impossible for other researchers to replicate the study under the same conditions or methodologies. 2. The experiment provides some essential information but omits significant details needed for replication, leading to considerable ambiguity in methodology or conditions. 3. The experiment design includes sufficient details for replication, but lacks clarity or completeness in certain areas, posing challenges for seamless reproducibility. 4. The experiment is well-documented with clear, detailed instructions and methodologies that allow for consistent replication, albeit with minor areas for improvement. 5. The experiment design is exemplary in its clarity, detail, and comprehensiveness, ensuring that other researchers can precisely and effortlessly replicate the study under identical conditions and methodologies.

Table 16: The examples of research idea generation results from the proposed full ResearchAgent.

Index	Types	Texts
1	Input	Title: Knowledge-Augmented Language Model Prompting for Zero-Shot Knowledge Graph Question Answering Abstract: Large Language Models (LLMs) are capable of performing zero-shot closed-book question answering tasks, based on their internal knowledge stored in parameters during pre-training. However, such internalized knowledge might be insufficient and incorrect, which could lead LLMs to generate factually wrong answers. Furthermore, fine-tuning LLMs to update their knowledge is expensive. To this end, we propose to augment the knowledge directly in the input of LLMs. Specifically, we first retrieve the relevant facts to the input question from the knowledge graph based on semantic similarities between the question and its associated facts. After that, we prepend the retrieved facts to the input question in the form of the prompt, which is then forwarded to LLMs to generate the answer. Our framework, Knowledge-Augmented language model PromptING (KAPING), requires no model training, thus completely zero-shot. We validate the performance of our KAPING framework on the knowledge graph question answering task, that aims to answer the user's question based on facts over a knowledge graph, on which ours outperforms relevant zero-shot baselines by up to 48% in average, across multiple LLMs of various sizes.
		Entities: ['Natural language', 'Learning management system', 'Question answering', 'Wikipedia', 'Artificial intelligence', 'Named-entity recognition', 'Quality assurance', 'Long short-term memory', 'English language', 'Google Books', 'Natural-language programming', 'LMS color space', 'GitHub', 'Logic learning machine', 'Machine translation', 'DBpedia', 'Integrated library system', 'Spanish language', 'Neural machine translation', 'ArXiv', 'Artificial general intelligence', 'Natural-language understanding', 'Nonlinear programming', 'Codex', 'LoRa', 'Adhesive', 'Common Crawl', 'ROUGE (metric)', 'Transformer', 'Tomas Mikolov']
	Problem	Developing a Multimodal Knowledge-Aware Prompting System for Multilingual Zero-Shot Question Answering across Structured and Unstructured Data Sources

Continued on the next page

Table 16 – Continued from the previous page

Index	Types	Texts
		The further refined and enhanced method for developing a multimodal knowledge-aware prompting system for multilingual zero-shot question answering (MMKP-ZSQA) incorporates advanced refinements in the following components to address the requirements for innovativeness, clarity, and generalizability: 1. Interconnected Multimodal Knowledge Orchestration (IMKO): - IMKO is an advanced evolution of SMKI, incorporating state-of-the-art language and vision models to handle data verbalization with context-aware disambiguation methods. - Techniques like attention mechanisms are employed to weigh the relevance of different data pieces when integrated, revealing how textual and visual data augment each other in a synthesized representation. 2. Linguistically Inclusive Retrieval Engine (LIRE): - LIRE extends EMKA with an emphasis on semantic understanding, using transformer-based models trained on diverse datasets, including idiomatic and cultural nuances across languages. - Specific algorithms to handle linguistic phenomena such as code-switching and transliteration are included, enhancing the application to a broader set of languages and dialects. 3. Prompt Learning and Optimization Nexus (PLON): - Building on AMPL, PLON focuses on creating a library of optimized prompts categorized by linguistic features and data modalities, using Bayesian optimization algorithms. - It includes domain adaptation techniques and a wider array of meta-learning strategies with case studies for high and low-resource languages, and outlines theoretical frameworks for their implementation. 4. Cross-Modal Integrative Learning System (C-MILS): - C-MILS advances SCAS by detailing the use of multi-head attention across modalities for effective and scalable reasoning, with exemplifications on how each modality can enhance comprehension synergistically. - The component now incorporates a layer of abstraction to distill knowledge into a modality-agnostic format, aiding reasoning and facilitating interpretability across languages and data types. 5. User-Centric Adaptation and Privacy Framework (UCAPF): - UCAPF enriches ALCUM with a user-centric design, focusing on interaction protocols that describe user engagement cycles, feedback loops, and privacy-preserving active learning paradigms, with practical workflows and pseudocode. - Aligns with GDPR and other privacy frameworks to fortify trust and address the ethical use of data in dynamic learning environments. 6. Global Evaluation and Reflective Testing System (GERTS): - GERTS expands CEB by presenting a multi-tiered validation and reflection methodology to adjust system components based on a diverse set of metrics, including fairness, bias, interpretability, and computational efficiency. - Offers a structured approach to cross-cultural evaluation, including the use of demographic and regional diversity in forming test cohorts.
	Method	
		The experiment, named "Refined Experiment for Multimodal Knowledge-Aware Prompting System for Multilingual Zero-shot Question Answering (RE-MKP-ZSQA)", aims to methodically develop and validate an advanced AI system. The experiment is streamlined to address feasibility, clarity, and reproducibility concerns while upholding robustness and validity by adhering to the following refined phases: 1. Detailed System Implementation Plan: - Provide a publicly accessible project roadmap with specific milestones, resource allocation, and timelines. 2. Dataset Curation with Clear Guidelines: - Publish precise annotation guidelines with strategies to prevent bias. - Document the dataset assembly process, including source selection and data processing procedures. 3. Transparent System Training: - Offer a detailed training protocol with hyperparameters, optimization strategies, and Bayesian optimization processes used in PLON. 4. Structured Zero-Shot Evaluation: - Outline evaluation metrics derived from GERTS with benchmark datasets to test zero-shot capabilities. 5. Clearer Interdisciplinary Evaluation Protocol: - Specify the composition of the evaluation committee, criteria for assessments, and methods for integrating the feedback. 6. Iterative Improvement with Validation Metrics: - Describe statistical methods for reflective assessment and continuous improvement, aligned with multi-tiered GERTS methodology. 7. User-Centric Design and Privacy Compliance Evaluation: - Structure user studies with targeted data points to assess usability and cultural adaptability. - Outline privacy compliance protocols to adhere to international standards. 8. Detailed Global Scalability Evaluation Method: - Define evaluation metrics for scalability tests and describe diverse infrastructural setups. 9. Enhanced Reporting for Reproducibility: - Commit to creating a comprehensive report with precise specifications, configurations, and instructions for replication purposes. - Utilize GitHub for version-controlled deposition of code and datasets, and arXiv for openly accessible experiment protocols and findings.
	Experiment	

Continued on the next page